Animating Rocks: Using AI to Teach Earth Science to Children Aged 6–9

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Abstract:

This study explores the impact of including AI-generated cartoon videos within an active, hands-on learning experience in the classroom, aiming to enhance K-2 students' ability to identify igneous rocks. After viewing an AI generated cartoon, students took a quick pop quiz that was included at the end of the video, testing their ability to identify the real life versions of the cartoon rocks in addition to other general facts. The students of each class recorded their answers on a separate sheet of paper, and their accuracy was compared to previously collected data surrounding the results of quizzes on inactively learned material. The exact results have yet to be documented at this time, however, our hypothesis is that the AI-generated cartoon will have a similar effectiveness of other active and hands-on learning techniques. Should this hypothesis be supported, it will continue to suggest that AI can be useful in incorporating engaging experiences that can enhance young learners' ability to identify geological materials. Though AI is not completely advanced enough totally generate a cartoon that requires no editing, its capacity to reach this level is possible. Further research and development into AI generated content that is both reliable and accurate before it can be permanently included in the active teaching approach.

Background:

This study is a continuation of a previous project that aimed to teach kindergarten and 2nd-grade students about Earth's natural processes through hands-on and engaging activities. Unlike the last study, tht aimed to focus on sedimentary rocks, this study focused on igneous rocks. The overall goal was to determine how effective AI generated cartoons in teaching children aged K-2. AI is a relatively new tool, and its effectiveness in an educational setting is currently being explored. We aimed to use AI to further expand our research in including an interactive and visual teaching style in the classroom, and see how well it held up in engaging students.

Figure 1: Screenshot of the editing process taking place in Kinemaster displaying how the clips are put together with transitions and sound. Rock on display in this figure is Basalt

Methodology:

This project included the use of AI generated cartoon images of rocks that were created by chat GPT, based on prompts that describe the physical characteristics of igneous rocks. The rocks in question were pumice, basalt, rhyolite, and obsidian. The images were then animated to do specific movements such as walking, jumping, and talking by the program hailuo.ai. These animations were short 6 second segments, however there were many of segments that were edited together with a text to speech voiceover to compose a ~5 minute video. Editing took place within the video editor, Kinemaster, where sound and text were added to the video. There was no specific plot, and the video simply composed of transitions to each rock where they would point out their main defining characteristics to the viewer.

After the cartoon portion was completed, a pop quiz was included at the end based on what was presented in the video. The viewer was met with 6 questions to which they had to identify some of the included rocks, answer true and false questions, and answering questions about other general facts about igneous rocks.

AI-Generated Rock Characters:

Figure 2: Screenshot of an AI-generated image of the igneous rock rhyolite (generated by ChatGPT) in a volcanic background taken from one of the 6 second video segments generated by Hailuo.ai. Followed by a screenshot of AI generated rocks (Also ChatGPT) pumice and obsidian.

Figure 3: Photo of the kindergarten class taken after the cardboard and playdoh activities, as they respond to the pop quiz seen in Figure 5.

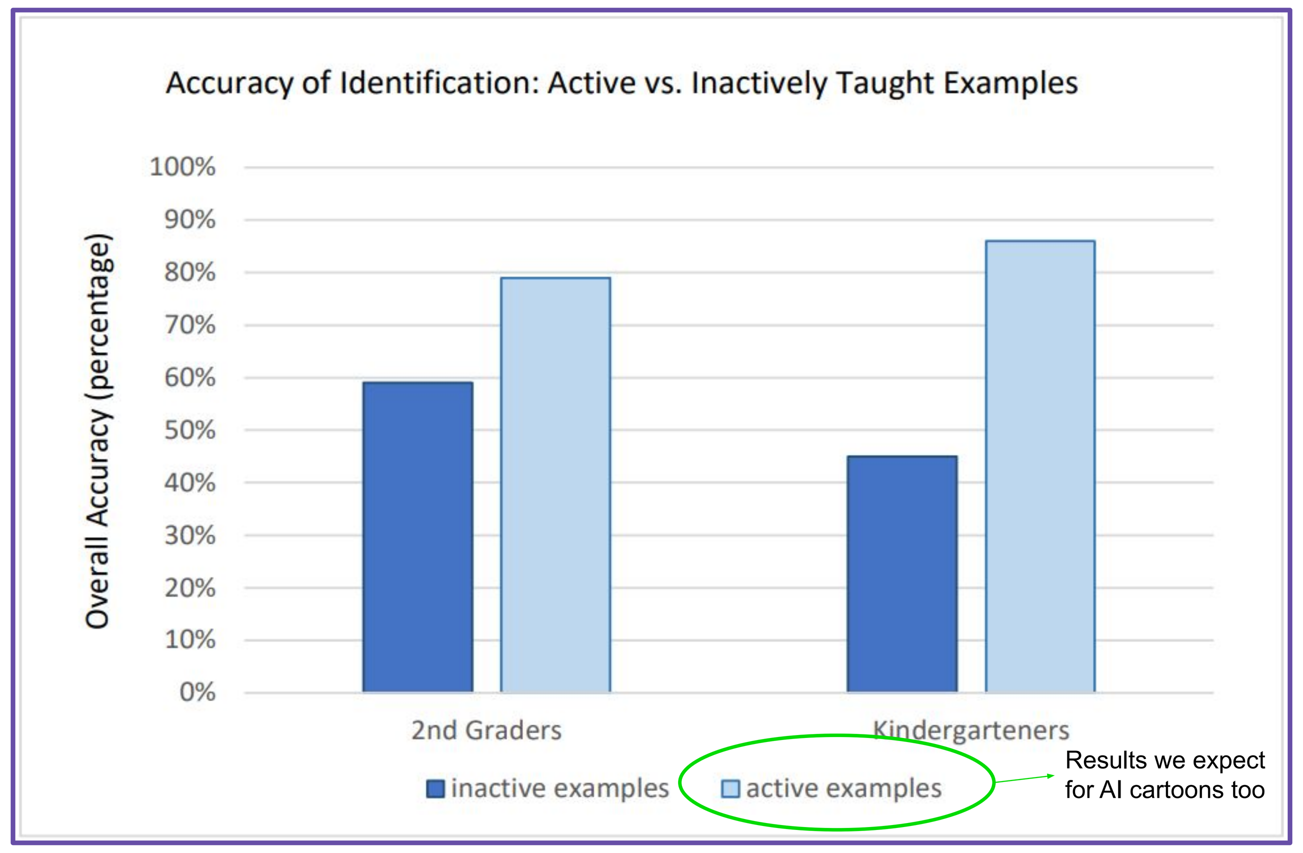

Figure 4: Bar graph comparing the students' sedimentary rock identification accuracies of rocks they were **actively** exposed to, and rocks that they were **inactively** exposed to.

This section contains research that we seek to link our future findings to. Figure 5 is a previously created graph showcasing the difference in student accuracy on quizzes based on whether they learned the material actively or inactively. The overall accuracy was 59% when identifying sedimentary rocks they were inactively exposed to, compared to 79% accuracy for rocks they were actively exposed to—showing a 20% difference between inactive and active examples. Should this graph be updated, we predict that our results for this study will reflect the trends seen in the accuracies of actively learned material, due to AI cartoons being a form of visual learning.

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Pop Quiz Questions:

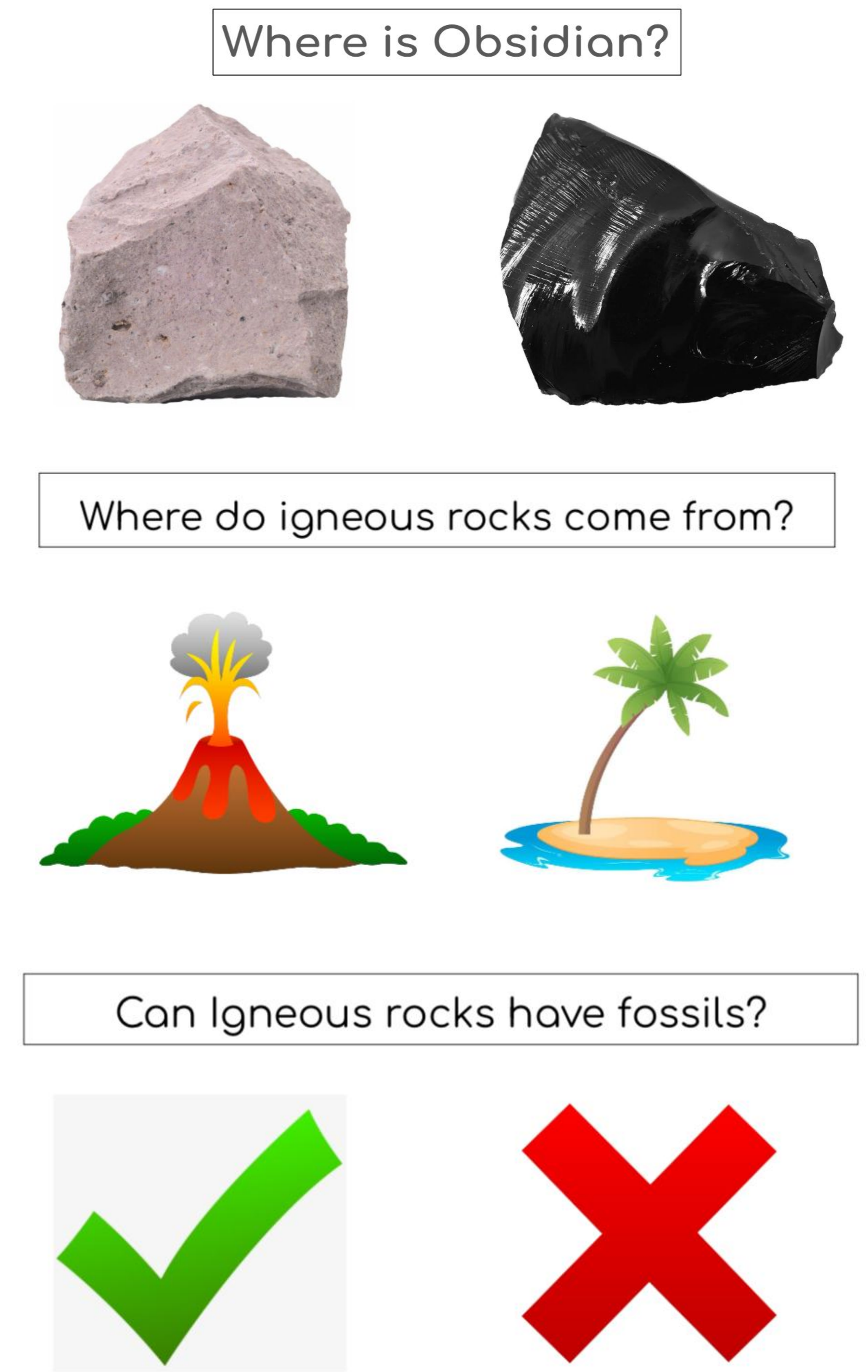

Figure 5: Photo of questions included in the pop quiz at the end of the cartoon section of the video. Due to the quiz being broadcasted on a big screen rather than individual papers, students will record their answers on a separate sheet of paper.

Discussion + Conclusion

Testing AI-generated igneous-rock cartoons, combined with hands-on activities, helped K–2 students recognize real rocks. After scoring the pop quizzes and comparing them with earlier sedimentary-rock lessons, where active learning produced much higher accuracy than inactive examples, we can see whether a similar pattern appears. If it does, this will support using short AI cartoons as a fun, visual way to strengthen early Earth-science learning.